Optimisation: gradients, gradient descent and its stochastic variants

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Training a model means minimising an objective over its parameters. This note reviews the calculus needed — partial derivatives, gradients, the Hessian — and derives gradient descent from a Taylor expansion: why the negative gradient is the steepest direction, why the learning rate must stay below $2/\lambda_{\max}$, and how conditioning sets the speed. It then treats convex and non-convex landscapes, the mini-batch stochastic gradients that make training on large datasets possible, and the refinements used in practice: momentum, Newton's method, Adam, line search and early stopping. Every method comes with pseudocode, its cost per iteration, and a small example computed by hand.

1 Learning is optimisation

The training phase of most algorithms in this course solves

$$\theta^* = \arg\min_{\theta \in \mathbb{R}^p} J(\theta), \qquad J(\theta) = \frac{1}{n}\sum_{i=1}^{n} L\big(f_\theta(\mathbf{x}_i), y_i\big) + \lambda\,\Omega(\theta),$$

the regularised empirical risk of Note 4, Section 7, as a function of the p parameters. Minimising the sum or the mean is the same problem, because multiplying a function by the positive constant $1/n$ does not move its minima; the mean keeps the scale of J, and hence a good learning rate, independent of n. Maximising a likelihood is the same as minimising its negative logarithm, so maximum-likelihood fitting (Note 2, Section 6) also fits this mould. Picture J as a landscape over parameter space: training means finding a low point. A note on letters: λ without an index is the regularisation strength (apart from the unknown of an eigenvalue equation $\det(H - \lambda I) = 0$), while λ_i, $\lambda_{\max}$ and $\lambda_{\min}$ are eigenvalues of the Hessian (Section 3.2).

The number of parameters ranges from $p = 2$ for a straight line $f(x) = wx + b$ through $p = d + 1$ for logistic regression on d features (a linear classifier with a smooth convex loss, Note 9, Section 4) to $p \approx 10^{11}$ for a large language model.

Symbols for running times. Throughout, n is the number of training examples, d the number of input features, p the number of parameters, n' the mini-batch size and T the number of iterations. We assume that the loss of one example and its gradient cost $O(p)$ operations; this holds for linear models ($p = d + 1$, one inner product) and, as the notes of week 5 on neural networks will show, for networks too, where one forward and backward pass costs a few operations per parameter. Hence one evaluation of J or of ∇J costs $O(np)$. An optimiser changes only the training cost: prediction with the trained model costs what the model costs ($O(p)$ per query for the models above), so the running-time boxes of this note list training time and memory.

When gradients are not available. Every method of this note needs continuous parameters and an objective whose gradient (or at least a subgradient) can be computed cheaply. Discrete choices — the k of k nearest neighbours, a histogram bin width, the splits of a tree, most hyperparameters — have no gradient; they are chosen by grid or random search on validation data (Note 4, Section 7). A piecewise-constant loss such as the 0–1 error has a gradient that is zero almost everywhere, so it is replaced by a smooth *surrogate* such as the logistic or hinge loss (Note 4, Section 2).

2 Derivatives, gradients and curvature

This section collects the calculus that every optimiser in the note relies on, using one small function as a running example.

2.1 Partial derivatives, the gradient and the Hessian

For one variable, the derivative $J'(\theta) = \lim_{\epsilon\to 0}\big(J(\theta+\epsilon) - J(\theta)\big)/\epsilon$ is the slope of the graph at θ. For $J : \mathbb{R}^p \to \mathbb{R}$ the same limit, taken along one coordinate axis, is the partial derivative

$$\frac{\partial J}{\partial \theta_k}(\theta) = \lim_{\epsilon\to 0} \frac{J(\theta + \epsilon\, \mathbf{e}_k) - J(\theta)}{\epsilon}$$

($\mathbf{e}_k$ is the k-th unit vector) measures the sensitivity of J to a change of θ_k alone, the other parameters held fixed; for $p = 1$ it is the ordinary derivative. In practice one differentiates with respect to θ_k treating the others as constants. The *gradient* collects the partial derivatives,

$$\nabla J(\theta) = \Big(\frac{\partial J}{\partial \theta_1}(\theta), \dots, \frac{\partial J}{\partial \theta_p}(\theta)\Big)^{\mathsf{T}} \in \mathbb{R}^p,$$

and the $p \times p$ matrix of second derivatives is the *Hessian*, $H_{kl} = \partial^2 J/\partial\theta_k \partial\theta_l$. For smooth J the order of differentiation does not matter, so H is symmetric and has real eigenvalues and orthogonal eigenvectors. Moving from θ along a vector $\mathbf{u}$, every coordinate changes as $\theta_k + \epsilon u_k$, with $d(\theta_k + \epsilon u_k)/d\epsilon = u_k$, so the chain rule gives the *directional derivative*

$$\frac{d}{d\epsilon} J(\theta + \epsilon\, \mathbf{u})\Big|_{\epsilon=0} = \sum_k \frac{\partial J}{\partial \theta_k}\, u_k = \nabla J(\theta)^{\mathsf{T}} \mathbf{u}. \tag{1}$$

The formula holds for any vector $\mathbf{u}$; for a unit vector it is the slope of J in that direction.

Worked example — gradient and Hessian of a two-parameter function

Take

$$J(\theta_1, \theta_2) = \tfrac{1}{4}\theta_1^4 - \theta_1\theta_2 + \tfrac{1}{2}\theta_2^2,$$

a small landscape that we shall reuse throughout the note. Differentiating with respect to one variable at a time,

$$\frac{\partial J}{\partial \theta_1} = \theta_1^3 - \theta_2, \qquad \frac{\partial J}{\partial \theta_2} = -\theta_1 + \theta_2, \qquad H(\theta) = \begin{pmatrix} 3\theta_1^2 & -1 \\ -1 & 1 \end{pmatrix},$$

where $H_{11} = \partial(\theta_1^3 - \theta_2)/\partial\theta_1 = 3\theta_1^2$, $H_{12} = \partial(\theta_1^3 - \theta_2)/\partial\theta_2 = -1$, $H_{21} = \partial(-\theta_1 + \theta_2)/\partial\theta_1 = -1$ (equal to H_{12}, as it must be) and $H_{22} = 1$. At $\theta = (2, 1)$:

$$J = 4 - 2 + \tfrac{1}{2} = 2.5, \qquad \nabla J = \begin{pmatrix} 8 - 1 \\ -2 + 1 \end{pmatrix} = \begin{pmatrix} 7 \\ -1 \end{pmatrix}, \qquad H = \begin{pmatrix} 12 & -1 \\ -1 & 1 \end{pmatrix}.$$

> The function rises steeply when θ_1 increases and falls slowly when θ_2 increases. The eigenvalues of $H(2,1)$ are 0.91 and 12.09: the surface curves upwards in every direction there, much more strongly along (roughly) the θ_1 axis.

Two gradients that recur. For a constant vector $\mathbf{b}$, $\mathbf{b}^\mathsf{T}\theta = \sum_k b_k\theta_k$ has $\partial/\partial\theta_k = b_k$, so $\nabla(\mathbf{b}^\mathsf{T}\theta) = \mathbf{b}$. For a symmetric matrix A, the quadratic form $\frac{1}{2}\theta^\mathsf{T}A\theta = \frac{1}{2}\sum_{i,j}\theta_i A_{ij}\theta_j$ has $\partial/\partial\theta_k = \frac{1}{2}\big(\sum_j A_{kj}\theta_j + \sum_i \theta_i A_{ik}\big) = (A\theta)_k$, using $A_{ik} = A_{ki}$; so $\nabla\big(\frac{1}{2}\theta^\mathsf{T}A\theta\big) = A\theta$, and its Hessian is A itself.

2.2 Taylor expansion and the direction of steepest descent

Taylor's theorem summarises what the gradient and the Hessian mean: for a small step $\boldsymbol{\delta}$,

$$J(\theta+\boldsymbol{\delta}) = J(\theta) + \nabla J(\theta)^\mathsf{T}\boldsymbol{\delta} + \tfrac{1}{2}\boldsymbol{\delta}^\mathsf{T}H(\theta)\,\boldsymbol{\delta} + O(\|\boldsymbol{\delta}\|^3). \tag{2}$$

The gradient gives the best linear approximation of J near θ, the Hessian the best quadratic correction.

Derivation: the negative gradient is the steepest descent direction. Take a step of small length ϵ in the direction of a unit vector $\mathbf{u}$, $\boldsymbol{\delta} = \epsilon\mathbf{u}$. Keeping the first-order term of (2),

$$J(\theta+\epsilon\mathbf{u}) - J(\theta) \approx \epsilon\,\nabla J(\theta)^\mathsf{T}\mathbf{u}.$$

Which unit vector makes this change as negative as possible? The Cauchy–Schwarz inequality states $|\mathbf{a}^\mathsf{T}\mathbf{b}| \le \|\mathbf{a}\|\,\|\mathbf{b}\|$, with equality exactly when $\mathbf{a}$ and $\mathbf{b}$ are parallel. With $\|\mathbf{u}\| = 1$ it gives

$$-\|\nabla J(\theta)\| \;\le\; \nabla J(\theta)^\mathsf{T}\mathbf{u} \;\le\; \|\nabla J(\theta)\|,$$

and the lower bound is reached only for $\mathbf{u} = -\nabla J(\theta)/\|\nabla J(\theta)\|$. Two conclusions follow. (i) The direction of *steepest descent* is $-\nabla J$; the gradient itself points uphill, in the direction of steepest ascent. (ii) The steepest slope equals $\|\nabla J\|$. In addition, (1) gives (iii): along a contour line J is constant, so $\nabla J^\mathsf{T}\mathbf{u} = 0$ for a tangent $\mathbf{u}$, and the gradient is perpendicular to the contour lines. These statements describe the slope at θ itself. A step of finite length follows them only approximately, because the gradient changes as θ moves; that is why they are called *local*, and why a long step in the steepest-descent direction can even increase J (Section 3).

In the worked example, at $(2,1)$ the steepest descent direction is $-(7,-1)/\sqrt{50} = (-0.990, 0.141)$, along which J falls at rate $\sqrt{50} = 7.07$; moving along $-\mathbf{e}_1$ instead it falls at rate 7, along $+\mathbf{e}_2$ at rate 1.

2.3 Stationary points and their classification

Derivation: at a minimum the gradient vanishes. Suppose θ^* is a local minimum but $\mathbf{g} = \nabla J(\theta^*) \neq \mathbf{0}$. Step a little against the gradient, $\boldsymbol{\delta} = -\epsilon\mathbf{g}$. By (2), $J(\theta^*+\boldsymbol{\delta}) = J(\theta^*) - \epsilon\,\|\mathbf{g}\|^2 + O(\epsilon^2)$, which is smaller than $J(\theta^*)$ for small enough $\epsilon > 0$: a contradiction. Hence every local minimum (of a differentiable J on all of $\mathbb{R}^p$) satisfies the *stationarity condition*

$$\nabla J(\theta^*) = \mathbf{0}, \qquad \text{i.e.} \quad \frac{\partial J}{\partial\theta_k}(\theta^*) = 0 \quad (k = 1,\dots,p).$$

The condition is necessary, not sufficient: maxima and saddle points satisfy it too.

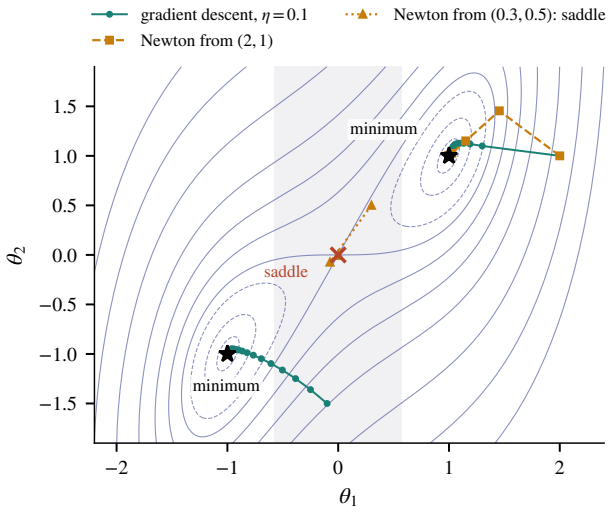


Figure 1. The landscape $J = \frac{1}{4}\theta_1^4 - \theta_1\theta_2 + \frac{1}{2}\theta_2^2$ with two minima (stars) and a saddle (cross). In the shaded strip $|\theta_1| < 1/\sqrt{3}$ the Hessian is not positive definite, so J is not convex (Section 4). Gradient descent with $\eta = 0.1$ from $(2, 1)$ and from $(-0.1, -1.5)$ reaches the nearest minimum. Newton's method (Section 6.2) from $(2, 1)$ reaches $(1, 1)$ in five steps to six digits, but from $(0.3, 0.5)$ it converges to the saddle.

The second-order test. At a stationary point the linear term of (2) vanishes and $J(\theta^* + \boldsymbol{\delta}) - J(\theta^*) \approx \frac{1}{2}\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta}$. Writing $H = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^{\mathsf{T}}$ with orthonormal eigenvectors $\mathbf{q}_i$,

$$\tfrac{1}{2}\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta} = \tfrac{1}{2} \sum_i \lambda_i \, (\mathbf{q}_i^{\mathsf{T}} \boldsymbol{\delta})^2 .$$

If all $\lambda_i > 0$ (H positive definite) every small step raises J: a local minimum. If all $\lambda_i < 0$ it is a local maximum. If some are positive and some negative, J rises along some directions and falls along others: a *saddle point*. Zero eigenvalues leave the test inconclusive. For $p = 2$ a quick rule uses $\det H = \lambda_1 \lambda_2$ and $\operatorname{tr} H = \lambda_1 + \lambda_2$: $\det H < 0$ means a saddle; $\det H > 0$ with $\operatorname{tr} H > 0$ a minimum; $\det H > 0$ with $\operatorname{tr} H < 0$ a maximum.

Worked example — finding and classifying the stationary points

For $J = \frac{1}{4}\theta_1^4 - \theta_1\theta_2 + \frac{1}{2}\theta_2^2$, set both partial derivatives to zero. The second gives $\theta_2 = \theta_1$; substituting into the first, $\theta_1^3 - \theta_1 = \theta_1(\theta_1 - 1)(\theta_1 + 1) = 0$. There are three stationary points:

point	J	H	$\det H$, $\operatorname{tr} H$	type
$(1, 1)$ and $(-1, -1)$	$-\frac{1}{4}$	$\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$	$2,\ 4$	minima (eigenvalues $2 \pm \sqrt{2} = 0.59,\ 3.41$)
$(0, 0)$	0	$\begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}$	$-1,\ 1$	saddle (eigenvalues $-0.62,\ 1.62$)

Figure 1 shows the landscape. Along the diagonal $\theta_1 = \theta_2 = s$ we have $J = \frac{1}{4}s^4 - \frac{1}{2}s^2$, which falls from the saddle towards both minima; along the anti-diagonal J rises.

Closed form or search. Sometimes $\nabla J = \mathbf{0}$ can be solved analytically, as in the example above. In machine learning, the class means of the nearest-centroid classifier (which assigns a point to the class with the closest mean, Note 9, Section 2), the maximum-likelihood Gaussian (Note 2, Section 6) and linear and ridge regression (least squares, without or with a penalty $\lambda \|\mathbf{w}\|^2$, Note 8) are such cases: their equations can be solved in closed form (for the means and the regression weights they

are even linear in the parameters). Usually they cannot be solved, and we must search. Trying all values on a grid (exhaustive search) costs G^p evaluations of J for G values per parameter, each $O(np)$, so $O(G^p\, np)$ in total. That is hopeless beyond a handful of parameters — ten values per parameter and $p = 20$ already give 10^{20} evaluations, the curse of dimensionality of Note 3 once more — so we follow the local slope instead. Grid and random search remain the method of choice for a few hyperparameters (Note 4, Section 7).

3 Gradient descent

Starting from an initial θ_0 (random, or by a heuristic suited to the model), repeat

$$\theta_{t+1} = \theta_t - \eta\, \nabla J(\theta_t), \tag{3}$$

where the *learning rate* (step size) $\eta > 0$ is a hyperparameter. By Section 2, each step moves in the locally steepest downhill direction, and for small enough η it decreases J: $J(\theta_{t+1}) \approx J(\theta_t) - \eta\, \|\nabla J(\theta_t)\|^2$.

Notation for iterates. The subscript $t = 0, 1, 2, \ldots$ on θ_t counts iterations. In the two-parameter examples, where θ_1 and θ_2 already denote the two components of θ, we write the iterates with a superscript instead, $\theta^{(t)} = (\theta_1^{(t)}, \theta_2^{(t)})$.

3.1 The algorithm and its cost

Algorithm 1 spells out batch gradient descent with the usual stopping rules: a small gradient norm, a negligible relative change of J, or an exhausted iteration budget. Stopping when the parameters hardly change is the gradient-norm test in disguise, since $\|\theta_{t+1} - \theta_t\| = \eta\, \|\nabla J(\theta_t)\|$ for a fixed η. One may also stop earlier, when the validation error stops improving (early stopping, Section 7).

Algorithm 1 Batch gradient descent with stopping rules

Require: objective J and its gradient ∇J; start θ_0; learning rate η; tolerances ϵ_g, ϵ_J; maximum T
Ensure: approximate minimiser θ

function GradientDescent(θ_0, η)
  $\theta \leftarrow \theta_0$, $J_{\text{old}} \leftarrow J(\theta)$
  for $t = 0, 1, \ldots, T-1$ **do**
    $\mathbf{g} \leftarrow \nabla J(\theta)$ ▷ mean of n per-example gradients: $O(np)$
    if $\|\mathbf{g}\| < \epsilon_g$ **then return** θ ▷ (nearly) stationary
    end if
    $\theta \leftarrow \theta - \eta\, \mathbf{g}$ ▷ $O(p)$; with constraints, $\Pi(\theta - \eta \mathbf{g})$ (Section 6.6)
    $J_{\text{new}} \leftarrow J(\theta)$ ▷ monitoring: $O(np)$; if J rises, reduce η
    if $|J_{\text{old}} - J_{\text{new}}| < \epsilon_J\, (1 + |J_{\text{old}}|)$ **then return** θ
    end if
    $J_{\text{old}} \leftarrow J_{\text{new}}$
  end for
  return θ ▷ iteration budget exhausted
end function

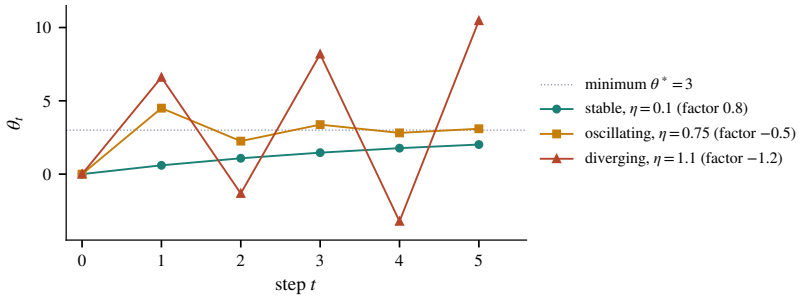


Figure 2. Iterates of gradient descent on $J(\theta) = (\theta - 3)^2$ from $\theta_0 = 0$ for three learning rates. The distance to the minimum is multiplied by $1 - 2\eta$ at every step.

Running time — batch gradient descent

Per iteration: one full gradient, $O(np)$ ($O(nd)$ for a linear model: one pass over the data matrix), plus $O(p)$ for the update. **Iterations:** on a quadratic with condition number κ (Section 3.2) and the best fixed step, about $\frac{\kappa}{2}\ln(1/\tau)$ to reduce the error by a factor τ (twice that for $\eta = 1/\lambda_{\max}$). **Memory:** $O(p)$ beyond the $O(nd)$ data. **Prediction:** unchanged by the optimiser, $O(p)$ per query. **Hardware:** the gradient is a matrix–vector product and vectorises well; the iterations themselves are sequential.

Worked example — gradient descent in one dimension: stable, oscillating, diverging

Let $J(\theta) = (\theta - 3)^2$, so $J'(\theta) = 2(\theta - 3)$, the minimum is $\theta^* = 3$ and the curvature is $J'' = 2$. The update (3) reads

$$\theta_{t+1} - 3 = (\theta_t - 3) - 2\eta(\theta_t - 3) = (1 - 2\eta)(\theta_t - 3),$$

so the distance to the minimum is multiplied by the same *factor* $1 - 2\eta$ at every step. From $\theta_0 = 0$:

η	factor	θ_1	θ_2	θ_3	θ_4	θ_5	$J(\theta_5)$	behaviour
0.1	0.8	0.6	1.08	1.464	1.771	2.017	0.966	slow, monotone
0.5	0	3	3	3	3	3	0	exact in one step
0.75	−0.5	4.5	2.25	3.375	2.813	3.094	0.009	oscillates, converges
1.1	−1.2	6.6	−1.32	8.184	−3.221	10.465	55.7	oscillates, diverges

Convergence requires $|1-2\eta| < 1$, i.e. $0 < \eta < 1 = 2/J''$. With a negative factor the iterate jumps across the minimum at every step; whether the jumps shrink or grow depends on whether $|1 - 2\eta|$ is below or above one (Figure 2). The best step, $\eta = 1/J'' = 0.5$, lands on the minimum at once: it is the Newton step of Section 6.2.

3.2 What the learning rate does on a quadratic

Near a minimum, (2) with $\boldsymbol{\delta} = \theta - \theta^*$ says that J is approximately quadratic,

$$J(\theta) \approx J^* + \tfrac{1}{2}(\theta - \theta^*)^{\mathsf{T}} A\,(\theta - \theta^*),$$

with $J^* = J(\theta^*)$ and $A = H(\theta^*)$ symmetric and positive definite; the linear term has dropped out because $\nabla J(\theta^*) = \mathbf{0}$. The behaviour of gradient descent on this model explains most of what one sees in practice.

Derivation: the per-direction factor. Write $\boldsymbol{\Delta} = \theta - \theta^*$ for the error. By the quadratic-form gradient of Section 2.1 (with $\boldsymbol{\Delta}$ in place of θ), $\nabla J(\theta) = A\boldsymbol{\Delta} = A(\theta - \theta^*)$. Substituting into (3),

$$\boldsymbol{\Delta}_{t+1} = \boldsymbol{\Delta}_t - \eta A \boldsymbol{\Delta}_t = (I - \eta A)\,\boldsymbol{\Delta}_t.$$

Expand the error in the orthonormal eigenvectors of A, $\boldsymbol{\Delta}_t = \sum_i z_{i,t}\,\mathbf{q}_i$ with $A\mathbf{q}_i = \lambda_i \mathbf{q}_i$ and coordinates $z_{i,t} = \mathbf{q}_i^{\mathsf{T}} \boldsymbol{\Delta}_t$ (here λ_i are eigenvalues, not the regularisation strength λ of Section 1). Because $(I - \eta A)\mathbf{q}_i = (1 - \eta\lambda_i)\mathbf{q}_i$, the coordinates decouple:

$$z_{i,t+1} = (1 - \eta\lambda_i)\, z_{i,t}, \qquad \text{so} \qquad z_{i,t} = (1 - \eta\lambda_i)^t\, z_{i,0}.$$

Along each eigenvector, gradient descent is exactly the one-dimensional example above, with curvature λ_i.

Derivation: the convergence condition. The error tends to zero from every start if and only if $|1 - \eta\lambda_i| < 1$ for all i, i.e. $0 < \eta\lambda_i < 2$. The most curved direction is the binding one:

$$\eta < \frac{2}{\lambda_{\max}}. \tag{4}$$

Per direction, $0 < \eta\lambda_i < 1$ gives a monotone approach, $\eta\lambda_i = 1$ one-step convergence, $1 < \eta\lambda_i < 2$ a damped oscillation across the valley and $\eta\lambda_i > 2$ an oscillation with growing amplitude.

Derivation: the best fixed step and the condition number. The slowest direction determines the overall speed. Because the $\mathbf{q}_i$ are orthonormal, $\|\boldsymbol{\Delta}_t\|^2 = \sum_i z_{i,t}^2$, and each z_i is multiplied by $|1 - \eta\lambda_i| \le \rho(\eta)$, where $\rho(\eta) = \max_i |1 - \eta\lambda_i| = \max(|1 - \eta\lambda_{\min}|,\, |1 - \eta\lambda_{\max}|)$. Hence $\|\boldsymbol{\Delta}_{t+1}\| \le \rho(\eta)\, \|\boldsymbol{\Delta}_t\|$: the error norm is *multiplied* by at most ρ per step. The first term decreases with η, the second increases once $\eta > 1/\lambda_{\max}$, so the best η balances them: $1 - \eta\lambda_{\min} = \eta\lambda_{\max} - 1$, giving

$$\eta_{\text{best}} = \frac{2}{\lambda_{\min} + \lambda_{\max}}, \qquad \rho_{\text{best}} = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1}, \qquad \kappa = \frac{\lambda_{\max}}{\lambda_{\min}},$$

where κ is the *condition number*. "Reducing the error by a factor τ" means multiplying it by $\tau < 1$. After t steps the error is at most ρ^t times its start, so $\rho^t = \tau$ needs $t = \ln(1/\tau)/\ln(1/\rho)$ steps. Writing $\rho_{\text{best}} = (1 - 1/\kappa)/(1 + 1/\kappa)$ and using $\ln(1+x) \approx x$ for small x, $\ln(1/\rho_{\text{best}}) = \ln(1 + 1/\kappa) - \ln(1 - 1/\kappa) \approx 2/\kappa$ for large κ, so about $\frac{\kappa}{2}\ln(1/\tau)$ steps are needed: elongated valleys are hard. For an error reduction of 10^{-6} this is 62 steps for $\kappa = 9$, 173 for $\kappa = 25$, 691 for $\kappa = 100$ and 69 078 for $\kappa = 10^4$. Only the ratio κ matters, not the size of the eigenvalues: multiplying J by a constant multiplies every λ_i by it, the best η changes by the inverse factor, and ρ_{best}, which depends only on κ, is unchanged.

Worked example — gradient descent on a two-dimensional quadratic

Let $J(\theta) = \frac{1}{2}\theta^{\mathsf{T}} A \theta$ with $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, i.e. $J = \frac{1}{2}(5\theta_1^2 + 8\theta_1\theta_2 + 5\theta_2^2)$, with minimum $\theta^* = \mathbf{0}$ and $\nabla J = (5\theta_1 + 4\theta_2,\ 4\theta_1 + 5\theta_2)$. From $\det(A - \lambda I) = (5 - \lambda)^2 - 16 = 0$ the eigenvalues are $\lambda_{\max} = 9$ with eigenvector along $(1, 1)$ and $\lambda_{\min} = 1$ along $(1, -1)$: $\kappa = 9$, the stability limit is $\eta < 2/9 = 0.222$ and the best fixed step is $\eta = 2/(1 + 9) = 0.2$ with $\rho = 0.8$.
Start at $\theta^{(0)} = (3, 1) = 2\,(1, 1) + 1\,(1, -1)$, where $J = 37$ and $\nabla J = (19, 17)$. With $\eta = 0.1$ the first step is $\theta^{(1)} = (3, 1) - 0.1\,(19, 17) = (1.1, -0.7)$. The eigen-decomposition predicts the same thing at a glance: the $(1, 1)$ part is multiplied by $1 - 0.1 \cdot 9 = 0.1$ and the $(1, -1)$ part by $1 - 0.1 \cdot 1 = 0.9$, so $\theta^{(t)} = 2\,(0.1)^t(1, 1) + (0.9)^t(1, -1)$. For three step sizes:

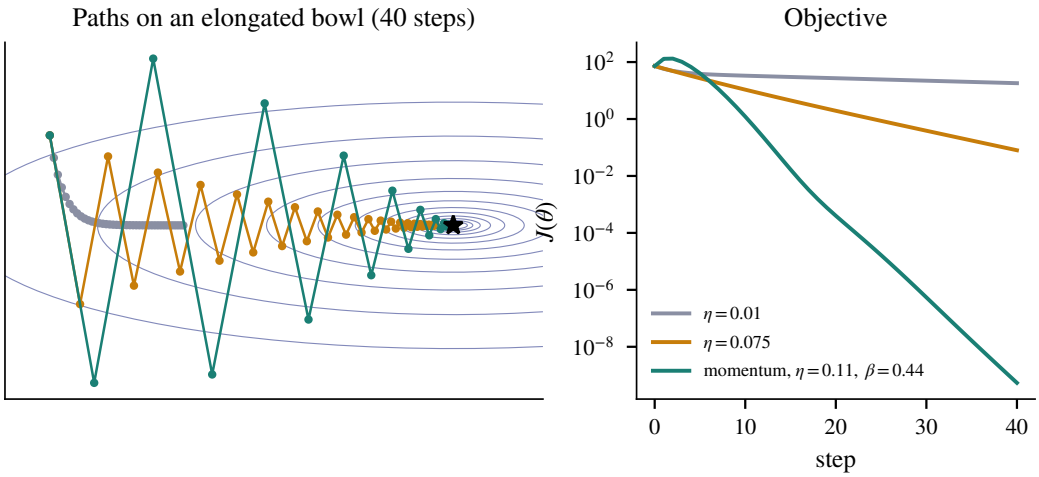


Figure 3. Forty steps on $J(\theta) = \frac{1}{2}(\theta_1^2 + 25\,\theta_2^2)$ ($\kappa = 25$) from the same start $(-9, 1.6)$. Left: horizontal axis θ_1, vertical axis θ_2 (the steep direction); the star is the minimum. After 40 steps $J = 18$ for $\eta = 0.01$, 0.08 for $\eta = 0.075$, and 5×10^{-10} with momentum (Section 6.1) using Polyak's choice for $\kappa = 25$, $\eta = 4/36 = 0.11$ and $\beta = (4/6)^2 = 0.44$.

η	factors $(1,1)$, $(1,-1)$	$\theta^{(1)}$	$\theta^{(2)}$	$\theta^{(3)}$	$J(\theta^{(1)})$, $J(\theta^{(2)})$, $J(\theta^{(3)})$
0.1	0.1, 0.9	$(1.1, -0.7)$	$(0.83, -0.79)$	$(0.731, -0.727)$	1.17, 0.660, 0.531
0.2	-0.8, 0.8	$(-0.8, -2.4)$	$(1.92, 0.64)$	$(-0.512, -1.536)$	23.7, 15.2, 9.70
0.25	-1.25, 0.75	$(-1.75, -3.25)$	$(3.69, 2.56)$	$(-3.48, -4.33)$	56.8, 88.2, 137.5

With $\eta = 0.1$ the steep direction is removed almost at once, but the shallow one shrinks only by 0.9 per step. With $\eta = 0.2$ the iterate jumps across the valley at every step (factor -0.8) and J looks worse at first; yet both directions now shrink by 0.8, and after 40 steps $J = 6.5 \times 10^{-7}$ against 2.2×10^{-4} for $\eta = 0.1$. With $\eta = 0.25 > 2/9$ the steep direction grows by 1.25 per step and the iteration diverges.

Figure 3 shows a valley with $\kappa = 25$. With $\eta = 0.01$ the descent is stable but crawls; with $\eta = 0.075$, close to the limit $2/25 = 0.08$, it zigzags across the valley and makes steady progress.

Why features are standardised. The curvatures λ_i come from the data. For the line fit $J(w, b) = \frac{1}{n}\sum_i \frac{1}{2}(wx_i + b - y_i)^2$ the first derivatives are $\frac{1}{n}\sum_i r_i x_i$ and $\frac{1}{n}\sum_i r_i$ with residuals $r_i = wx_i + b - y_i$; differentiating once more ($\partial r_i/\partial w = x_i$, $\partial r_i/\partial b = 1$) gives the Hessian as the matrix of averages $\begin{pmatrix} \overline{x^2} & \bar{x} \\ \bar{x} & 1 \end{pmatrix}$. With inputs $x = 1, 2, 3, 4$ it is $\begin{pmatrix} 7.5 & 2.5 \\ 2.5 & 1 \end{pmatrix}$, with eigenvalues 8.35 and 0.150 and $\kappa = 56$; after centring the inputs $(x - 2.5)$ it becomes $\mathrm{diag}(1.25, 1)$ with $\kappa = 1.25$. Very different feature scales, or features far from zero mean, stretch the valley; standardising them (Note 5, Section 5) is often the cheapest way to speed up gradient descent.

3.3 Learning-rate schedules

A fixed step is rarely ideal: large steps make fast early progress, small steps settle into the minimum, and with stochastic gradients (Section 5) a decreasing step is needed to converge at all. Three common schedules, all with peak value $\eta_{\max}$:

- $1/t$ decay: $\eta_t = \eta_{\max}\, t_0/(t_0 + t)$, halving after t_0 steps;

Algorithm 2 Gradient step with backtracking (Armijo) line search

Require: current θ; $\mathbf{g} = \nabla J(\theta)$; initial step η_0; $c \in (0,1)$ (e.g. 10^{-4} to 0.5); shrink factor $\frac{1}{2}$

function BacktrackingStep($\theta, \mathbf{g}$)
  $\eta \leftarrow \eta_0$
  while $J(\theta - \eta\mathbf{g}) > J(\theta) - c\,\eta\,\|\mathbf{g}\|^2$ **do** ▷ each test: one evaluation of J
    $\eta \leftarrow \eta/2$
  end while
  return $\theta - \eta\mathbf{g}$
end function

- step decay: multiply η by 0.1 every fixed number of steps (or when the validation loss stalls);
- linear warm-up over the first W steps, then cosine decay up to step T: $\eta_t = \eta_{\max}\,(t+1)/W$ for $t < W$, then $\eta_t = \frac{1}{2}\eta_{\max}\big(1 + \cos(\pi\,(t-W)/(T-W))\big)$ for $W \le t \le T$ [10].

The linear *warm-up* from a small value avoids instabilities while the parameters are still random and the curvature is unknown; the half-cosine decays slowly at first and approaches zero smoothly at the end (some recipes stop the decay at a floor such as $0.1\,\eta_{\max}$).

Worked example — three schedules side by side

With $\eta_{\max} = 0.1$, $T = 1000$, $t_0 = 100$, step decay every 400 steps and a warm-up of $W = 100$ steps:

step t	0	50	100	325	550	775	1000
$1/t$ decay	0.1000	0.0667	0.0500	0.0235	0.0154	0.0114	0.0091
step decay	0.1	0.1	0.1	0.1	0.01	0.01	0.001
warm-up + cosine	0.0010	0.0510	0.1000	0.0854	0.0500	0.0146	0

For instance at $t = 325$ the cosine schedule gives $0.05\,(1+\cos(\pi\cdot 225/900)) = 0.05\,(1+\cos 45^\circ) = 0.0854$.

3.4 Choosing the step automatically: backtracking line search

When the full gradient is available, the step can be chosen at each iteration instead of fixed in advance. By (2), a step $-\eta\mathbf{g}$ should lower J by about $\eta\,\|\mathbf{g}\|^2$ if η is small. The *Armijo condition* [1] accepts η if it achieves at least a fraction $c \in (0,1)$ of this predicted decrease,

$$J(\theta - \eta\mathbf{g}) \le J(\theta) - c\,\eta\,\|\mathbf{g}\|^2 ;$$

backtracking starts from a generous η and halves it until the condition holds (Algorithm 2). A small η always passes eventually, because the first-order prediction becomes exact as $\eta \to 0$ and $c < 1$.

Worked example — one backtracking step

For $J(\theta) = (\theta - 3)^2$ at $\theta = 0$: $J = 9$, $g = -6$, $g^2 = 36$; take $\eta_0 = 1$, $c = \frac{1}{4}$. Trial $\eta = 1$: $J(6) = 9 > 9 - \frac{1}{4}\cdot 1 \cdot 36 = 0$, rejected (the step overshoots to the mirror point). Trial $\eta = \frac{1}{2}$: $J(3) = 0 \le 9 - \frac{1}{4}\cdot\frac{1}{2}\cdot 36 = 4.5$, accepted; it lands on the minimum.

Running time — backtracking line search

Per iteration: one gradient, $O(np)$, plus one evaluation of J, $O(np)$, per trial (usually one or two). **Memory:** $O(p)$. It needs exact values of J; with noisy mini-batch estimates the test is unreliable, which is why deep learning uses schedules instead.

4 Convex and non-convex objectives

A function is *convex* if the chord between any two points of its graph lies on or above the graph:

$$J\big(\alpha\mathbf{a} + (1-\alpha)\mathbf{b}\big) \le \alpha J(\mathbf{a}) + (1-\alpha)J(\mathbf{b}) \qquad \text{for all } \mathbf{a}, \mathbf{b} \text{ and } 0 \le \alpha \le 1.$$

For twice differentiable J this is equivalent to $H(\theta)$ being positive semi-definite everywhere: the surface curves upwards (or is flat) in every direction, at every point.

Derivation: for a convex function, stationary means global minimum. Put $\alpha = 1 - s$ in the definition, with small $s > 0$: $J(\mathbf{a} + s(\mathbf{b} - \mathbf{a})) \le J(\mathbf{a}) + s\,(J(\mathbf{b}) - J(\mathbf{a}))$. Subtract $J(\mathbf{a})$, divide by s and let $s \to 0$; by (1) with $\mathbf{u} = \mathbf{b} - \mathbf{a}$ the left side becomes the directional derivative, so

$$J(\mathbf{b}) \ge J(\mathbf{a}) + \nabla J(\mathbf{a})^{\mathsf{T}}(\mathbf{b} - \mathbf{a}) \qquad \text{for all } \mathbf{b} \ :$$

the tangent plane lies below the graph everywhere. If $\nabla J(\mathbf{a}) = \mathbf{0}$, then $J(\mathbf{b}) \ge J(\mathbf{a})$ for every $\mathbf{b}$, so $\mathbf{a}$ is a global minimum. Hence every local minimum of a convex function is global, and, when a minimiser exists, gradient descent with a suitable step approaches the global minimum value from any start. (A minimiser need not exist: unregularised logistic regression on separable data keeps lowering its loss as the weights grow without bound, Note 9, Section 4.)

Sums of convex functions and positive multiples of convex functions are convex, so a mean of convex per-example losses plus a convex penalty such as $\lambda \|\theta\|^2$ is convex. Least squares, ridge regression, logistic regression and the linear support vector machine (a linear classifier that maximises the margin between the classes; Notes 8 and 9) all have convex objectives for this reason. The quadratic of the previous worked example is convex (eigenvalues 1 and 9, the same at every point). The landscape of Figure 1 is not: $H = \begin{pmatrix} 3\theta_1^2 & -1 \\ -1 & 1 \end{pmatrix}$ has $\det H = 3\theta_1^2 - 1$, which is negative in the strip $|\theta_1| < 1/\sqrt{3} \approx 0.577$, and two separate minima can only exist because of it. For a convex J, two minimisers $\mathbf{a}$ and $\mathbf{b}$ have the same (global) value J^*, and the chord inequality gives $J \le J^*$, hence $J = J^*$, at every point between them: the minimisers of a convex function form one connected set, never two isolated points.

Neural networks are not convex either. Their landscapes have many stationary points, and where gradient descent ends depends on where it starts (Figures 1 and 4). In high dimension the picture is less bleak than these low-dimensional cartoons suggest: most stationary points with high loss are saddle points rather than poor local minima, because it is unlikely that all p curvatures are positive at once, and local minima tend to have loss close to that of the global minimum [6]. Slow progress near saddles, where the gradient is small, rather than being trapped, is the more common difficulty; a small perturbation, such as mini-batch noise, pushes gradient descent off a saddle that has a direction of negative curvature, along that direction.

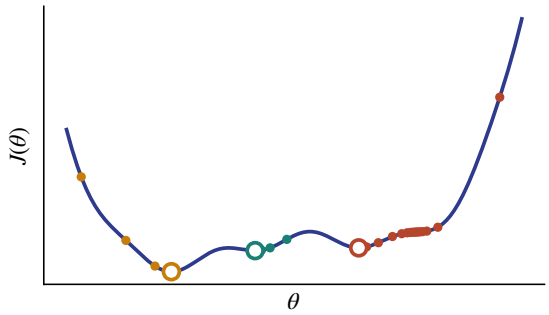


Figure 4. Gradient descent ($\eta = 0.02$, 200 steps) from $\theta_0 = -2.8$, 0 and 2.9 on a non-convex function ends in three different local minima (open circles), at $\theta = -1.57, -0.43$ and 0.98; only the first is the global minimum.

5 Stochastic gradients

5.1 The gradient of a mean

Derivation: the gradient of a mean is the mean of the gradients. Differentiation is linear: the derivative of a sum is the sum of the derivatives and constants factor out. Applying this to each component $\partial/\partial\theta_k$ of $J(\theta) = \frac{1}{n}\sum_i L_i(\theta)$, where $L_i(\theta) = L(f_\theta(\mathbf{x}_i), y_i)$,

$$\nabla J(\theta) = \nabla\Big(\frac{1}{n}\sum_{i=1}^{n} L_i(\theta)\Big) = \frac{1}{n}\sum_{i=1}^{n} \nabla L_i(\theta) \quad (+\,\lambda\nabla\Omega).$$

Batch gradient descent computes all n terms before every update, which costs a full pass over the data per step: for $n = 10^6$ images, one step means processing a million images. *Mini-batch* gradient descent replaces the mean over D_{train} by the mean over a random subset B of $n' \ll n$ examples:

$$\mathbf{g}_B(\theta) = \frac{1}{n'}\sum_{i\in B} \nabla L_i(\theta), \qquad \theta \leftarrow \theta - \eta\,\mathbf{g}_B(\theta).$$

The case $n' = 1$ is *stochastic* (online) gradient descent, SGD; $n' = n$ recovers the batch method.

5.2 Mini-batch gradients are unbiased and their noise falls like $1/n'$

Let $\bar{\mathbf{g}} = \nabla J(\theta) = \frac{1}{n}\sum_i \nabla L_i$ and let $\Sigma = \frac{1}{n}\sum_i (\nabla L_i - \bar{\mathbf{g}})(\nabla L_i - \bar{\mathbf{g}})^{\mathsf{T}}$ be the covariance of the per-example gradients over the training set (both at the current θ).

Derivation: unbiasedness. Draw an index I uniformly from $\{1, \dots, n\}$. Then $\mathbb{E}[\nabla L_I] = \sum_i \frac{1}{n}\nabla L_i = \bar{\mathbf{g}}$. A mini-batch of n' draws $I_1, \dots, I_{n'}$ gives $\mathbf{g}_B = \frac{1}{n'}\sum_k \nabla L_{I_k}$, and by linearity of expectation

$$\mathbb{E}[\mathbf{g}_B] = \frac{1}{n'}\sum_{k=1}^{n'} \mathbb{E}[\nabla L_{I_k}] = \bar{\mathbf{g}}.$$

On average, the mini-batch gradient points exactly where the full gradient points. (The same holds for batches drawn without replacement, since each example is then equally likely to be in B.)

Derivation: variance. Each draw ∇L_{I_k} is a uniformly chosen per-example gradient, so its covariance is Σ by definition. If the n' draws are independent (with replacement), the covariance of a sum

of independent vectors is the sum of their covariances, and a constant factor a multiplies a covariance by a^2, so

$$\operatorname{Cov}[\mathbf{g}_B] = \frac{1}{n'^2}\sum_{k=1}^{n'} \operatorname{Cov}[\nabla L_{I_k}] = \frac{1}{n'^2}\, n'\,\Sigma = \frac{\Sigma}{n'}.$$

Each component's standard deviation falls like $1/\sqrt{n'}$: quadrupling the batch halves the noise, while it quadruples the cost of a step. Without replacement the covariance is smaller by the factor $(n - n')/(n-1)$, which vanishes for $n' = n$, as it should.

Worked example — a mini-batch gradient from a table of four examples

Fit $f(x) = wx + b$ with $L_i = \frac{1}{2}(wx_i + b - y_i)^2$. With residual $r_i = wx_i + b - y_i$ the chain rule gives $\partial L_i/\partial w = r_i x_i$ and $\partial L_i/\partial b = r_i$. At $(w, b) = (1, 0)$:

i	x_i	y_i	$f(x_i)$	r_i	$\partial L_i/\partial w = r_i x_i$	$\partial L_i/\partial b = r_i$
1	1	2	1	-1	-1	-1
2	2	3	2	-1	-2	-1
3	3	5	3	-2	-6	-2
4	4	6	4	-2	-8	-2
mean (full gradient)					-4.25	-1.5

Here $J = \frac{1}{4}(0.5{+}0.5{+}2{+}2) = 1.25$. The mini-batch $B = \{1, 3\}$ gives $\mathbf{g}_B = \frac{1}{2}\big((-1,-1)+(-6,-2)\big) = (-3.5, -1.5)$: same signs, different size. The six possible batches of two give

$$\{1,2\} : (-1.5,-1), \quad \{1,3\} : (-3.5,-1.5), \quad \{1,4\} : (-4.5,-1.5),$$
$$\{2,3\} : (-4,-1.5), \quad \{2,4\} : (-5,-1.5), \quad \{3,4\} : (-7,-2),$$

whose average is $(-4.25, -1.5)$, exactly the full gradient: unbiasedness. The w-component of a single-example gradient has variance 8.19 over the four examples; for batches of two drawn with replacement it is $8.19/2 = 4.09$, and for the six pairs above (without replacement) $4.09 \times (4-2)/(4-1) = 2.73$. One update with $\eta = 0.05$ and $B = \{1, 3\}$ moves (w, b) from $(1, 0)$ to $(1 + 0.175,\ 0 + 0.075) = (1.175, 0.075)$ and lowers the full objective from 1.25 to 0.544. (The least-squares solution is $(1.4, 0.5)$; the stability limit from the Hessian eigenvalue 8.35 is $\eta < 0.24$.)

5.3 Epochs, shuffling and the noise floor

In practice the data are shuffled once per *epoch* (one pass over D_{train}) and cut into consecutive mini-batches (Algorithm 3). Every example is then used exactly once per epoch, and the shuffling prevents systematic patterns: data sorted by class would push the parameters back and forth between classes for whole stretches of an epoch.

Running time — mini-batch SGD

Per update: $O(n'p)$; with $\lceil n/n' \rceil$ updates per epoch, one epoch costs $O(np)$, the same as *one* batch gradient step. **Total:** $O(Enp)$ for E epochs. **Memory:** $O(p)$ plus per-batch intermediate values ($\propto n'$); the data can be streamed from disk. **Hardware:** the n' gradients are computed together as matrix products, so on a GPU a batch of 256 costs far less than 256 times one example (for small models, hardly more than one example).

Figure 5 compares the three on logistic regression with $n = 2000$. After one epoch, batch gradient descent has taken one step; mini-batch descent with $n' = 32$ has taken 63 and is already within 0.013 of the optimal loss. With a constant step, however, the stochastic methods do not converge: they

Algorithm 3 Mini-batch stochastic gradient descent with shuffling

Require: training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$; start θ_0; batch size n'; schedule η_t; number of epochs E; clipping threshold G (optional)
Ensure: parameters θ

function MiniBatchSGD(θ_0)
  $\theta \leftarrow \theta_0, \quad t \leftarrow 0$
  for epoch $= 1, \dots, E$ **do**
    $\pi \leftarrow$ random permutation of $\{1, \dots, n\}$ ▷ $O(n)$
    for $j = 0, n', 2n', \dots$ while $j < n$ **do**
      $B \leftarrow \{\pi_{j+1}, \dots, \pi_{\min(j+n', n)}\}$
      $\mathbf{g} \leftarrow \frac{1}{|B|} \sum_{i \in B} \nabla_\theta L\big(f_\theta(\mathbf{x}_i), y_i\big)$ ▷ $O(n'p)$, vectorised
      optional clipping: **if** $\|\mathbf{g}\| > G$ **then** $\mathbf{g} \leftarrow G\,\mathbf{g}/\|\mathbf{g}\|$ ▷ $O(p)$; Section 7
      $\theta \leftarrow \theta - \eta_t\,\mathbf{g}, \quad t \leftarrow t + 1$
    end for
    optionally evaluate the validation loss (Section 7)
  end for
  return θ
end function

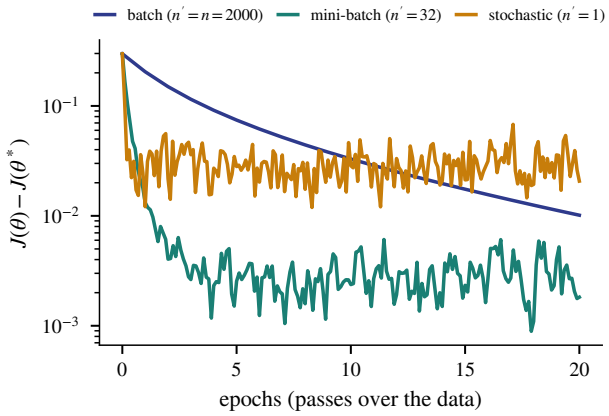


Figure 5. Excess training loss of logistic regression ($d = 20$, $n = 2000$) against the number of passes over the data, with constant learning rates. Mini-batch descent reaches 0.0018 after 20 epochs; batch descent 0.010; SGD fluctuates between about 0.015 and 0.07 over the last epochs (0.02 at the end).

fluctuate in a band whose width is set by the gradient noise and the learning rate. The reason is visible in the derivation above: at the minimum the full gradient vanishes, but the per-example gradients do not ($\Sigma \neq 0$), so each step still moves θ by about η times the noise. Pure SGD, the noisiest, ends further from the optimum than batch descent; mini-batches are the compromise used in practice.

Decreasing the learning rate removes the noise floor at the end of training. Robbins and Monro [16] gave the classical conditions on the schedule,

$$\sum_t \eta_t = \infty, \qquad \sum_t \eta_t^2 < \infty \ :$$

the steps must add up to an unlimited distance, so that any starting point can be left behind, while the accumulated noise, whose variance grows like $\sum_t \eta_t^2$, stays finite. The $1/t$ schedule of Section 3.3 satisfies both; a constant step violates the second, and $\eta_t = 1/t^2$ violates the first.

6 Beyond plain gradient descent

6.1 Momentum

Accumulate a velocity and move with it:

$$\mathbf{v}_{t+1} = \beta\,\mathbf{v}_t - \eta \nabla J(\theta_t), \qquad \theta_{t+1} = \theta_t + \mathbf{v}_{t+1}, \qquad 0 \le \beta < 1,$$

with $\mathbf{v}_0 = \mathbf{0}$; $\beta = 0$ is plain gradient descent. Unrolling the recursion, $\mathbf{v}_{t+1} = -\eta\big(\mathbf{g}_t + \beta\mathbf{g}_{t-1} + \beta^2\mathbf{g}_{t-2} + \cdots\big)$: the step is an exponentially weighted sum of past gradients. If the gradient keeps the same value $\mathbf{g}$, the velocity tends to $-\eta\mathbf{g}\,(1 + \beta + \beta^2 + \cdots) = -\eta\mathbf{g}/(1-\beta)$, an effective learning rate ten times larger for $\beta = 0.9$. If it alternates in sign, $\pm\mathbf{g}$, the sum alternates, $1 - \beta + \beta^2 - \cdots = 1/(1+\beta)$, and its size tends to $\eta\,\|\mathbf{g}\|\,/(1+\beta)$: components of the gradient that keep their sign add up, those that flip, as across a valley, cancel. As $\beta \to 1$ the friction vanishes: the effective step $\eta/(1-\beta)$ grows without bound, and the iterate overshoots and oscillates for a long time before it settles.

On a quadratic, Polyak's choice $\beta = \big((\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)\big)^2$ and $\eta = 4/(\sqrt{\lambda_{\max}} + \sqrt{\lambda_{\min}})^2$ improves the convergence factor per step from $(\kappa-1)/(\kappa+1)$ to $(\sqrt{\kappa}-1)/(\sqrt{\kappa}+1)$ [15]. The number of steps then grows like $\sqrt{\kappa}$ instead of κ: for an error reduction of 10^{-6}, 20 steps instead of 62 for $\kappa = 9$, 35 instead of 173 for $\kappa = 25$, and 691 instead of 69 078 for $\kappa = 10^4$. In Figure 3 momentum reaches $J \approx 5 \times 10^{-10}$ in forty steps. Nesterov's accelerated gradient [12], a variant that evaluates the gradient at the look-ahead point $\theta_t + \beta\mathbf{v}_t$ (one changed line in Algorithm 4), provably needs only about $\sqrt{\kappa}\,\ln(1/\tau)$ iterations for all smooth, strongly convex functions (Hessian eigenvalues bounded below by a positive constant everywhere), not only for quadratics.

Algorithm 4 Gradient descent with momentum (heavy ball)

Require: start θ_0; learning rate η; momentum $\beta \in [0, 1)$ (typically 0.9); number of steps T
Ensure: approximate minimiser θ

```
function Momentum(θ_0)
    θ ← θ_0,   v ← 0
    for t = 0, ..., T − 1 do
        g ← ∇J(θ)                          ▷ or g_B; Nesterov: g ← ∇J(θ + βv)
        v ← βv − ηg                        ▷ O(p)
        θ ← θ + v
    end for
    return θ
end function
```

Running time — momentum

Per iteration: the gradient ($O(np)$ batch, $O(n'p)$ mini-batch) plus $O(p)$. **Memory:** one extra vector $\mathbf{v}$. **Iterations:** about $\frac{\sqrt{\kappa}}{2}\ln(1/\tau)$ on a quadratic with tuned η, β, against $\frac{\kappa}{2}\ln(1/\tau)$ without momentum — for a negligible extra cost. Nesterov's variant costs exactly the same: one gradient plus $O(p)$.

Worked example — two momentum steps

Return to $J = \frac{1}{2}\theta^{\mathsf{T}} A\theta$ with $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\theta^{(0)} = (3, 1)$, and take $\eta = 0.2$, $\beta = 0.25$.

Step 1. $\mathbf{g}_0 = (19, 17)$, $\mathbf{v}_1 = 0.25 \cdot \mathbf{0} - 0.2\,(19, 17) = (-3.8, -3.4)$, $\theta^{(1)} = (-0.8, -2.4)$, $J = 23.7$ — the same as plain gradient descent, since the velocity starts at zero.

Step 2. $\mathbf{g}_1 = A\theta^{(1)} = (-13.6, -15.2)$, $\mathbf{v}_2 = 0.25\,(-3.8, -3.4) - 0.2\,(-13.6, -15.2) = (-0.95 +$

$2.72,\ -0.85+3.04) = (1.77, 2.19)$, $\theta^{(2)} = (-0.8+1.77,\ -2.4+2.19) = (0.97, -0.21)$, $J = 1.65$. Plain gradient descent with the same η has $J = 15.2$ after two steps: the old velocity $(-3.8, -3.4)$ points against the new descent step $-\eta\mathbf{g}_1 = (2.72, 3.04)$ along the steep direction $(1, 1)$, so the overshoot is partly cancelled. Over more steps:

method	$J(\theta^{(1)})$	$J(\theta^{(2)})$	$J(\theta^{(10)})$	$J(\theta^{(20)})$	error factor
gradient descent, $\eta = 0.1$	1.17	0.660	0.122	0.0148	0.9
gradient descent, $\eta = 0.2$	23.7	15.2	0.427	0.0049	0.8
momentum, $\eta = 0.2$, $\beta = 0.25$	23.7	1.65	0.0010	5.0×10^{-7}	0.69
momentum, $\eta = 0.25$, $\beta = 0.25$	56.8	36.2	0.0088	3.2×10^{-8}	0.5

The last row is Polyak's choice for $\kappa = 9$: $\beta = (2/4)^2 = 0.25$, $\eta = 4/(3+1)^2 = 0.25$, factor 0.5. Note that $\eta = 0.25$ alone diverges (previous worked example); the momentum term stabilises it. The last column gives the asymptotic factor by which the *error* shrinks per step, so J shrinks by its square. For momentum it is the largest $|r|$ among the roots of $r^2 - (1 + \beta - \eta\lambda_i)\, r + \beta = 0$ over both eigenvalues (the eigenvalues of the 2×2 iteration matrix of Question C3 with $a = \lambda_i$); for $\eta = 0.2$, $\beta = 0.25$ and $\lambda_i = 1$ the roots are 0.685 and 0.365, for $\lambda_i = 9$ a complex pair of modulus $\sqrt{\beta} = 0.5$. Polyak's row starts with a larger overshoot and a slower transient, and overtakes the momentum row with $\eta = 0.2$ only from step 15 on.

6.2 Newton's method

Derivation: the Newton step. Instead of the first-order model, minimise the second-order model of (2) around the current point,

$$m(\boldsymbol{\delta}) = J(\theta) + \mathbf{g}^{\mathsf{T}}\boldsymbol{\delta} + \tfrac{1}{2}\boldsymbol{\delta}^{\mathsf{T}} H \boldsymbol{\delta}, \qquad \mathbf{g} = \nabla J(\theta),\ H = H(\theta).$$

Its gradient with respect to $\boldsymbol{\delta}$ is $\mathbf{g} + H\boldsymbol{\delta}$ (Section 3.2, with H symmetric). Setting it to zero gives $H\boldsymbol{\delta} = -\mathbf{g}$, so

$$\theta_{t+1} = \theta_t - H(\theta_t)^{-1}\, \nabla J(\theta_t),$$

and the stationary point of m is its minimum when H is positive definite. In one dimension this is $\theta \leftarrow \theta - J'(\theta)/J''(\theta)$: Newton's root-finding method applied to J'. Compared with gradient descent, the Hessian rescales every eigen-direction by its own curvature, $1/\lambda_i$, which is the per-direction step $\eta\lambda_i = 1$ that converged in one step in the one-dimensional example. Conditioning no longer matters.

Worked example — Newton steps

On the quadratic $J = \frac{1}{2}\theta^{\mathsf{T}} A \theta$ with $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, the model m is J itself. From $\theta^{(0)} = (3, 1)$: $A^{-1} = \frac{1}{9}\begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}$ and $A^{-1}\mathbf{g}_0 = \frac{1}{9}(5 \cdot 19 - 4 \cdot 17,\ -4 \cdot 19 + 5 \cdot 17) = \frac{1}{9}(27, 9) = (3, 1)$, so $\theta^{(1)} = (3, 1) - (3, 1) = (0, 0)$: the exact minimum in one step, whatever the condition number.

On the non-quadratic landscape $J = \frac{1}{4}\theta_1^4 - \theta_1\theta_2 + \frac{1}{2}\theta_2^2$ from $(2, 1)$: $\mathbf{g} = (7, -1)$ and $H = \begin{pmatrix} 12 & -1 \\ -1 & 1 \end{pmatrix}$ with $\det H = 11$, so $H^{-1} = \frac{1}{11}\begin{pmatrix} 1 & 1 \\ 1 & 12 \end{pmatrix}$ and $H^{-1}\mathbf{g} = \frac{1}{11}(7 - 1,\ 7 - 12) = (\frac{6}{11}, -\frac{5}{11})$. The new point is $\theta^{(1)} = (2 - \frac{6}{11},\ 1 + \frac{5}{11}) = (\frac{16}{11}, \frac{16}{11}) = (1.4545, 1.4545)$. Repeating, the iterates stay on the diagonal and approach the minimum $(1, 1)$:

step t	0	1	2	3	4	5
$\theta_1^{(t)} - 1$	1	0.455	0.151	0.0253	9.1×10^{-4}	1.2×10^{-6}

Once close, the error is roughly squared at each step (the number of correct digits doubles): *quadratic convergence*. Gradient descent with $\eta = 0.1$ from the same start needs 86 steps to bring both coordinates within 10^{-3} of the minimum and about 200 to match the accuracy Newton reaches in five (Figure 1). From $(0.3, 0.5)$, however, Newton's method jumps to $(-0.074, -0.074)$ and then converges to the saddle $(0, 0)$, where H is indefinite: it looks for stationary points, not specifically for minima.

The remedies for this behaviour are standard [13]: add μI to H until it is positive definite (a *Hessian modification*, the idea behind the Levenberg–Marquardt method for least squares), which makes $\boldsymbol{\delta}$ a descent direction; take a *damped* step $\alpha\boldsymbol{\delta}$ with α chosen by line search; or restrict the step to a region where the quadratic model can be trusted (trust-region methods). Algorithm 5 combines the first two. With it, the run from $(0.3, 0.5)$ uses $\mu = 0.512$ and $\alpha = \frac{1}{4}$ in its first step and reaches the minimum $(1, 1)$, to eight digits, in three steps instead of the saddle.

Algorithm 5 Damped Newton's method with Hessian modification

Require: start θ_0; gradient ∇J and Hessian $H(\theta)$; tolerance ϵ_g; maximum T; Armijo constant $c \in (0, 1)$ (e.g. 10^{-4})
Ensure: approximate minimiser θ

function NEWTON(θ_0)
  $\theta \leftarrow \theta_0$
  for $t = 0, \dots, T-1$ **do**
    $\mathbf{g} \leftarrow \nabla J(\theta)$ ▷ $O(np)$
    if $\|\mathbf{g}\| < \epsilon_g$ **then return** θ
    end if
    $H \leftarrow H(\theta)$, $\mu \leftarrow 0$ ▷ $O(np^2)$ to form, $O(p^2)$ memory
    while the Cholesky factorisation of $H + \mu I$ fails **do** ▷ $O(p^3)$ per attempt
      $\mu \leftarrow \max(2\mu, 10^{-3})$ ▷ until $H + \mu I$ is positive definite
    end while
    solve $(H + \mu I)\boldsymbol{\delta} = -\mathbf{g}$ with the factor ▷ $O(p^2)$; never form H^{-1}
    $\alpha \leftarrow 1$ ▷ $\mathbf{g}^{\mathsf{T}}\boldsymbol{\delta} < 0$: a descent direction
    while $J(\theta + \alpha\boldsymbol{\delta}) > J(\theta) + c\,\alpha\,\mathbf{g}^{\mathsf{T}}\boldsymbol{\delta}$ **do** $\alpha \leftarrow \alpha/2$
    end while
    ▷ Armijo test of Section 3.4 with $\boldsymbol{\delta}$ in place of $-\mathbf{g}$
    $\theta \leftarrow \theta + \alpha\boldsymbol{\delta}$
  end for
  return θ
end function

Running time — Newton's method

Per iteration: gradient $O(np)$; Hessian $O(np^2)$ (for logistic regression $H = \frac{1}{n}X^{\mathsf{T}}SX$, with X the $n \times (d+1)$ data matrix and S a diagonal matrix of per-example weights, Note 9, Section 4: $O(nd^2)$); solving the $p \times p$ system $O(p^3)$. **Memory:** $O(p^2)$. **Iterations:** typically 5 to 20. For p up to a few thousand this is an excellent trade, and logistic regression is often trained this way. In 32-bit floats the Hessian needs 4 TB for $p = 10^6$ and 40 PB for $p = 10^8$ (the parameters: 0.4 GB). Mini-batches do not rescue it: a Hessian

estimated from a mini-batch is noisy, and inverting it amplifies the noise along its smallest eigenvalues, so steps can explode. Stochastic training therefore uses cheap diagonal rescalings instead, such as Adam's division by $\sqrt{\hat{\mathbf{s}}}$ (Section 6.4).

Between the two extremes lie methods that use curvature without storing H; we name them for orientation only, their details are beyond this course [13]. *Quasi-Newton* methods such as L-BFGS approximate H^{-1} by a scaled identity corrected with M rank-two updates built from the last M (typically 5–20) differences of gradients and parameters, at $O(Mp)$ extra time and memory per step. The *conjugate gradient* method, which minimises a quadratic in at most p steps using only products of the Hessian with vectors, is simple enough to treat in full.

6.3 Conjugate gradient

Take the quadratic $J(\theta) = \frac{1}{2}\theta^{\mathsf{T}} A\theta - \mathbf{b}^{\mathsf{T}}\theta$ with A symmetric positive definite (least squares has this form, Note 8), whose minimiser solves $A\theta = \mathbf{b}$. The residual $\mathbf{r} = \mathbf{b} - A\theta = -\nabla J$ is the steepest-descent direction. The conjugate gradient method [8] moves along directions $\mathbf{d}_t$ that start from the residual but are corrected by a multiple of the previous direction so that they are *A-conjugate*, $\mathbf{d}_s^{\mathsf{T}} A\mathbf{d}_t = 0$ for $s \neq t$, and along each one it takes the exact minimising step: setting $\frac{d}{d\alpha} J(\theta + \alpha\mathbf{d}) = -\mathbf{r}^{\mathsf{T}}\mathbf{d} + \alpha\, \mathbf{d}^{\mathsf{T}} A\mathbf{d}$ to zero gives $\alpha = \mathbf{r}^{\mathsf{T}}\mathbf{d} / \mathbf{d}^{\mathsf{T}} A\mathbf{d}$, which equals $\mathbf{r}^{\mathsf{T}}\mathbf{r} / \mathbf{d}^{\mathsf{T}} A\mathbf{d}$ for these directions. Conjugate directions do not undo each other's progress, so in exact arithmetic the minimum is reached in at most p steps; in practice the error falls by a factor τ in about $\frac{\sqrt{\kappa}}{2}\ln(2/\tau)$ steps [13], the same $\sqrt{\kappa}$ as tuned momentum but without having to know $\lambda_{\min}$ and $\lambda_{\max}$. For a non-quadratic J, *nonlinear* conjugate gradient (Fletcher–Reeves, Polak–Ribière) uses $-\nabla J$ for $\mathbf{r}$ and a line search for α.

Algorithm 6 Linear conjugate gradient for $A\theta = \mathbf{b}$

Require: symmetric positive definite A (or a routine computing $A\mathbf{u}$); $\mathbf{b}$; start θ_0; tolerance ϵ_g
Ensure: minimiser θ of $\frac{1}{2}\theta^{\mathsf{T}} A\theta - \mathbf{b}^{\mathsf{T}}\theta$

function ConjugateGradient(θ_0)
  $\theta \leftarrow \theta_0, \quad \mathbf{r} \leftarrow \mathbf{b} - A\theta, \quad \mathbf{d} \leftarrow \mathbf{r}$
  for $t = 0, \dots, p-1$ **do**
    if $\|\mathbf{r}\| < \epsilon_g$ **then return** θ
    end if
    $\mathbf{u} \leftarrow A\mathbf{d}$ ▷ the only product with A
    $\alpha \leftarrow \mathbf{r}^{\mathsf{T}}\mathbf{r} / \mathbf{d}^{\mathsf{T}}\mathbf{u}$ ▷ exact minimum along $\mathbf{d}$
    $\theta \leftarrow \theta + \alpha\mathbf{d}, \quad \mathbf{r}_{\text{new}} \leftarrow \mathbf{r} - \alpha\mathbf{u}$ ▷ $\mathbf{r}_{\text{new}} = \mathbf{b} - A\theta$
    $\beta \leftarrow \mathbf{r}_{\text{new}}^{\mathsf{T}}\mathbf{r}_{\text{new}} / \mathbf{r}^{\mathsf{T}}\mathbf{r}, \quad \mathbf{d} \leftarrow \mathbf{r}_{\text{new}} + \beta\mathbf{d}, \quad \mathbf{r} \leftarrow \mathbf{r}_{\text{new}}$ ▷ $O(p)$
  end for
  return θ
end function

Running time — conjugate gradient

Per iteration: one product $A\mathbf{d}$ plus $O(p)$: $O(p^2)$ for a dense A, and $O(np)$ for least squares, where $A\mathbf{d} = \frac{1}{n} X^{\mathsf{T}}(X\mathbf{d})$ is computed with two matrix–vector products without ever forming A. **Memory:** four vectors, $O(p)$, beyond the data. **Iterations:** at most p in exact arithmetic, about $\frac{\sqrt{\kappa}}{2}\ln(2/\tau)$ in practice. It needs exact gradients, so like line search it is a method for deterministic, full-batch problems.

Worked example — conjugate gradient on the running quadratic

For $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\mathbf{b} = \mathbf{0}$ and $\theta^{(0)} = (3, 1)$: $\mathbf{r}_0 = \mathbf{d}_0 = (-19, -17)$, $\mathbf{u} = A\mathbf{d}_0 = (-163, -161)$, $\mathbf{r}_0^{\mathsf{T}}\mathbf{r}_0 = 650$, $\mathbf{d}_0^{\mathsf{T}}\mathbf{u} = 5834$, so $\alpha_0 = 0.111$ and $\theta^{(1)} = (0.883, -0.894)$ with $J = 0.790$ (gradient descent with $\eta = 0.1$: 1.17). The new residual $\mathbf{r}_1 = (-0.839, 0.938)$ is orthogonal to $\mathbf{r}_0$; $\beta_0 = 1.584/650 = 0.00244$, $\mathbf{d}_1 = (-0.886, 0.897)$ and $\alpha_1 = 0.997$ give $\theta^{(2)} = (0, 0)$ exactly. With $p = 2$, two steps suffice; Newton's method needs only one, but it must solve a linear system with A, whereas conjugate gradient only multiplies vectors by A.

6.4 Adaptive methods: Adam and AdamW

Adam [9] keeps exponential moving averages of the gradient and of its element-wise square,

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1)\, \mathbf{g}_t, \qquad \mathbf{s}_t = \beta_2 \mathbf{s}_{t-1} + (1 - \beta_2)\, \mathbf{g}_t^2,$$

with $\mathbf{m}_0 = \mathbf{s}_0 = \mathbf{0}$, and moves each parameter by its averaged gradient divided by the root mean square of its recent gradients:

$$\theta_t = \theta_{t-1} - \eta \, \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{s}}_t} + \epsilon}, \qquad \hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \quad \hat{\mathbf{s}}_t = \frac{\mathbf{s}_t}{1 - \beta_2^t},$$

all operations element-wise; typical values are $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$.

Derivation: the bias correction. Unrolling the recursion from $\mathbf{m}_0 = \mathbf{0}$, $\mathbf{m}_t = (1-\beta_1)\sum_{j=1}^{t} \beta_1^{t-j} \mathbf{g}_j$. If all gradients were equal to $\mathbf{g}$, the geometric sum would give $\mathbf{m}_t = (1 - \beta_1)\, \frac{1-\beta_1^t}{1-\beta_1}\, \mathbf{g} = (1 - \beta_1^t)\, \mathbf{g}$, which is too small at the start because the average was initialised at zero. Dividing by $1 - \beta_1^t$ removes this bias; the same argument applies to $\mathbf{s}_t$. After many steps $\beta^t \to 0$ and the correction disappears.

The ratio $\hat{\mathbf{m}}/\sqrt{\hat{\mathbf{s}}}$ is dimensionless and of order one in size: $\hat{\mathbf{m}}$ is an average of recent gradients and $\sqrt{\hat{\mathbf{s}}}$ their root mean square, and the size of an average never exceeds the root mean square (their squares differ by a variance, which is non-negative). With equal weights in the two averages the ratio is at most 1, with equality when the gradient keeps a constant sign and size; with $\beta_1 \neq \beta_2$ this holds approximately. So the step of each parameter is typically of size η (and bounded by a small multiple of it), whatever the scale of its gradient. This compensates for very different gradient scales across the layers of a network, the per-parameter analogue of standardising the features. The price: the step no longer shrinks as the gradient shrinks, so near a minimum Adam keeps moving by about η unless η is decayed (Section 3.3), and it stores two extra vectors.

Worked example — Adam's first steps

Let $J = \frac{1}{2}(\theta_1^2 + 25\,\theta_2^2)$, so $\mathbf{g} = (\theta_1, 25\theta_2)$, start at $(2, 1)$ and use $\eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ (ϵ negligible).
Step 1. $\mathbf{g}_1 = (2, 25)$; $\mathbf{m}_1 = 0.1\,\mathbf{g}_1 = (0.2, 2.5)$; $\mathbf{s}_1 = 0.001\,\mathbf{g}_1^2 = (0.004, 0.625)$; the corrections divide by 0.1 and 0.001, so $\hat{\mathbf{m}}_1 = (2, 25)$ and $\hat{\mathbf{s}}_1 = (4, 625)$. The step is $0.1\,(2/2,\ 25/25) = (0.1, 0.1)$ and $\theta^{(1)} = (1.9, 0.9)$. The first Adam step is η sign$(\mathbf{g})$.
Step 2. $\mathbf{g}_2 = (1.9, 22.5)$; $\mathbf{m}_2 = 0.9\,(0.2, 2.5) + 0.1\,(1.9, 22.5) = (0.37, 4.5)$; $\mathbf{s}_2 = 0.999\,(0.004, 0.625) + 0.001\,(3.61, 506.25) = (0.007606, 1.1306)$. With $1 - 0.9^2 = 0.19$ and $1 - 0.999^2 = 0.001999$: $\hat{\mathbf{m}}_2 = (1.947, 23.68)$, $\hat{\mathbf{s}}_2 = (3.805, 565.6)$, $\sqrt{\hat{\mathbf{s}}_2} = (1.951, 23.78)$, and the step is $(0.0998, 0.0996)$, so $\theta^{(2)} = (1.800, 0.800)$.

Algorithm 7 Adam and AdamW

Require: start θ_0; learning rates η_t; $\beta_1, \beta_2 \in [0, 1)$; $\epsilon > 0$; weight decay $\lambda \geq 0$ ($\lambda = 0$: Adam); number of steps T

Ensure: parameters θ

1: **function** AdamW(θ_0)
2: $\theta \leftarrow \theta_0, \quad \mathbf{m} \leftarrow \mathbf{0}, \quad \mathbf{s} \leftarrow \mathbf{0}$
3: **for** $t = 1, \dots, T$ **do**
4: $\mathbf{g} \leftarrow \mathbf{g}_B(\theta)$ ▷ mini-batch gradient, $O(n'p)$
5: $\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1 - \beta_1)\mathbf{g}, \quad \mathbf{s} \leftarrow \beta_2 \mathbf{s} + (1 - \beta_2)\, \mathbf{g} \odot \mathbf{g}$ ▷ $\odot$: element-wise product
6: $\hat{\mathbf{m}} \leftarrow \mathbf{m}/(1 - \beta_1^t), \quad \hat{\mathbf{s}} \leftarrow \mathbf{s}/(1 - \beta_2^t)$
7: $\theta \leftarrow \theta - \eta_t\big(\hat{\mathbf{m}}/(\sqrt{\hat{\mathbf{s}}} + \epsilon) + \lambda\, \theta\big)$ ▷ element-wise; $\lambda\theta$: decoupled decay
8: **end for**
9: **return** θ
10: **end function**

> Both coordinates move by almost exactly η, although their gradients differ by a factor of 12. Plain gradient descent must keep $\eta < 2/25 = 0.08$ for stability; with $\eta = 0.07$ its first step moves the component θ_1 by only 0.14 and θ_2 by 1.75, overshooting to $(1.86, -0.75)$.

Weight decay: Adam versus AdamW. Adding $\frac{\lambda}{2}\|\theta\|^2$ to J adds $\lambda\theta$ to the gradient. In Adam this term is then divided by $\sqrt{\hat{\mathbf{s}}}$, so parameters with large gradients are barely regularised. AdamW [11] instead *decouples* the decay, subtracting $\eta_t \lambda \theta$ directly in the update (line 7 of Algorithm 7), so every parameter shrinks at the same relative rate. (This is the convention of common libraries such as PyTorch, where the decay is scaled by the learning rate; the original paper scales λ by the schedule multiplier only, so the same nominal λ can mean different things.) For plain gradient descent the two are identical; for adaptive methods they are not, and AdamW is the default optimiser for transformers.

Running time — Adam and AdamW

Per iteration: the mini-batch gradient, $O(n'p)$, plus $O(p)$ element-wise operations. **Memory:** $\mathbf{m}$ and $\mathbf{s}$ double the model size; with parameters and gradient, about $4p$ numbers — 1.6 GB for $p = 10^8$ in 32-bit floats, against 40 PB for a Hessian. **Prediction:** unchanged; $\mathbf{m}, \mathbf{s}$ are discarded.

6.5 Where the gradients come from

Every method above needs ∇J. The definition of the partial derivative suggests *finite differences*: the forward difference $\big(J(\theta + h\mathbf{e}_k) - J(\theta)\big)/h$ has error $O(h)$, the central difference

$$\frac{\partial J}{\partial \theta_k}(\theta) \approx \frac{J(\theta + h\mathbf{e}_k) - J(\theta - h\mathbf{e}_k)}{2h}$$

has error $O(h^2)$, because the second-order Taylor terms cancel: by (2), $J(\theta \pm h\mathbf{e}_k) = J(\theta) \pm h\, \partial J/\partial\theta_k + \frac{1}{2}h^2 H_{kk} + O(h^3)$, so the difference is $2h\, \partial J/\partial\theta_k + O(h^3)$. The step cannot be made arbitrarily small: rounding errors in the difference of two nearly equal numbers grow like $1/h$; in double precision $h \approx 10^{-5}$ is a good compromise for the central difference. For $J = \frac{1}{4}\theta_1^4 - \theta_1\theta_2 + \frac{1}{2}\theta_2^2$ at $(2, 1)$, where the exact gradient is $(7, -1)$, the forward differences give $(7.620, -0.950)$ for $h = 0.1$ and $(7.060, -0.995)$ for $h = 0.01$; the central differences give $(7.020, -1.000)$ and $(7.0002, -1.0000)$: dividing h by ten divides the forward error by ten and the central error by a hundred.

Algorithm 8 Gradient by central differences

Require: function J; point $\theta \in \mathbb{R}^p$; step h

function FiniteDifferenceGradient(θ, h)
  for $k = 1, \dots, p$ **do**
    $g_k \leftarrow \big(J(\theta + h\mathbf{e}_k) - J(\theta - h\mathbf{e}_k)\big)/(2h)$ ▷ two evaluations of J
  end for
  return $\mathbf{g} = (g_1, \dots, g_p)$
end function

Algorithm 9 Projected gradient descent

Require: gradient ∇J; projection Π onto the feasible set; feasible start θ_0; learning rate η; number of steps T

Ensure: feasible approximate minimiser θ

function ProjectedGD(θ_0)
  $\theta \leftarrow \theta_0$
  for $t = 0, \dots, T-1$ **do**
    $\mathbf{g} \leftarrow \nabla J(\theta)$ ▷ $O(np)$
    $\theta \leftarrow \Pi(\theta - \eta\mathbf{g})$ ▷ gradient step, then back to the nearest feasible point
  end for
  return θ
end function

Running time — finite differences versus automatic differentiation

Finite differences: $2p$ evaluations of J, each $O(np)$, so $O(np^2)$ per gradient; for $p = 10^6$ that is two million passes over the data for a single step. **Reverse-mode automatic differentiation** (backpropagation) computes the exact gradient at a cost of a small constant times one evaluation of J, $O(np)$, independently of p [2]; its price is memory, since the intermediate values of the forward computation are stored for the backward pass. Finite differences remain useful for *gradient checking*: comparing a hand-coded or automatic gradient with Algorithm 8 on a few random coordinates catches most bugs. How reverse mode achieves its cost is the subject of the week 5 notes on neural networks.

6.6 Constraints

Parameters may have to satisfy constraints: a variance must be positive, a spin direction must have unit length, the weights of a mixture must form a probability vector (non-negative, summing to one). One can reparameterise (write $\sigma^2 = e^u$, or obtain the mixture weights from free parameters u_k through the softmax $w_k = e^{u_k} / \sum_j e^{u_j}$, which is positive and sums to one), use Lagrange multipliers, or project. *Projected gradient descent* (Algorithm 9) changes a single line of Algorithm 1: $\theta \leftarrow \Pi(\theta - \eta\mathbf{g})$, where Π returns the nearest feasible point. For non-negativity Π sets negative components to zero; for a unit-length direction it divides by the norm.

Running time — projected gradient descent

Per iteration: the gradient, $O(np)$, plus the projection: $O(p)$ for non-negativity, a box or a unit sphere, $O(p \log p)$ for the probability simplex (by sorting). **Memory:** $O(p)$. **Iterations:** for a convex J and a convex feasible set, of the same order as unconstrained gradient descent.

Worked example — projected gradient descent with a non-negativity constraint

Minimise $J = (\theta_1 + 1)^2 + (\theta_2 - 2)^2$ subject to $\theta_1, \theta_2 \geq 0$, with $\eta = 0.25$ from $(1, 1)$; $\nabla J = (2(\theta_1 + 1),\ 2(\theta_2 - 2))$. Step 1: $(1, 1) - 0.25\,(4, -2) = (0, 1.5)$, already feasible. Step 2: $(0, 1.5) - 0.25\,(2, -1) = (-0.5, 1.75)$, projected to $(0, 1.75)$. Step 3 gives $(0, 1.875)$. The iterates approach $(0, 2)$ with $J = 1$, a point on the boundary; the unconstrained minimum $(-1, 2)$ is infeasible. At the constrained minimum the gradient $(2, 0)$ does not vanish: it points into the feasible region, and $-\nabla J$ points out of it, so no feasible step can lower J.

When the objective and all constraints are linear the problem is a *linear programme*; a quadratic objective with linear constraints is a *quadratic programme*. Both have dedicated solvers (simplex and interior-point methods [4]), and the support vector machine of Note 9, Section 5, is a quadratic programme.

In physics — optimisers as mechanics

Gradient descent with small steps follows the *gradient flow* $\dot{\theta} = -\nabla J(\theta)$: overdamped motion of a particle in the potential J. The update (3) is the explicit Euler discretisation of this flow with time step η, and the condition $\eta < 2/\lambda_{\max}$ is the familiar stability limit of explicit Euler for a stiff system: the fastest mode sets the largest stable time step. Momentum restores inertia — Polyak's "heavy ball", $m\ddot{\theta} + \gamma\dot{\theta} = -\nabla J$, discretised — so progress along the floor of a narrow valley builds up speed, while oscillations across it are damped; with Polyak's tuning they still occur but shrink by a factor $\sqrt{\beta}$ per step (Figure 3). The noise of mini-batch gradients acts like a temperature: SGD resembles Langevin dynamics, and adding properly scaled Gaussian noise to the updates turns it into a sampler of the posterior distribution of the parameters [17].

7 Stopping, monitoring and the cost of training

7.1 Early stopping

The stopping rules of Algorithm 1 look at the training objective. For learning, what matters is the error on new data, and that can start rising while the training loss still falls: the model begins to fit the noise of the training set (Note 4, Section 5). *Early stopping* monitors the loss on a validation set (Note 4, Section 4) after every epoch, remembers the best parameters seen, and stops once the validation loss has not improved for a number of epochs called the *patience* (Algorithm 10). The number of steps then acts like a regularisation hyperparameter: started from small weights, gradient descent on least squares stopped after few steps behaves much like ridge regression (least squares plus a penalty $\lambda \|\mathbf{w}\|^2$, Note 8). Along a low-curvature direction J barely changes, so the data determine the minimiser poorly there and noise in the data moves it a lot; by Section 3.2, gradient descent fits exactly these directions last (factor $1 - \eta\lambda_i$ close to one), so stopping early leaves them near their small starting values, much as the ridge penalty does (see also [7], Section 7.8).

Worked example — tracing early stopping

Validation losses after epochs $1, \ldots, 9$: $0.92, 0.61, 0.48, 0.44, 0.43, 0.45, 0.44, 0.47, 0.50$; $P = 3$. The best improves up to epoch 5 (0.43). Epoch 6 sets the counter to 1; epoch 7 (0.44) beats epoch 6 but not the best, so the counter becomes 2; epoch 8 makes it $3 = P$. Training stops and returns epoch 5's parameters; epoch 9 is never run.

Algorithm 10 Early stopping with patience

Require: initial θ; one-epoch training routine (e.g. the inner loop of Algorithm 3); validation set D_{valid} of n_{valid} examples; patience P; maximum epochs E

function EarlyStopping(θ)
  $\theta_{\text{best}} \leftarrow \theta, \quad \ell_{\text{best}} \leftarrow \infty, \quad k \leftarrow 0$ ▷ k: epochs without improvement
  for epoch $= 1, \dots, E$ **do**
    $\theta \leftarrow$ TrainOneEpoch(θ) ▷ $O(np)$
    $\ell \leftarrow$ validation loss of f_θ on D_{valid} ▷ $O(n_{\text{valid}}\, p)$
    if $\ell < \ell_{\text{best}}$ **then**
      $\ell_{\text{best}} \leftarrow \ell, \quad \theta_{\text{best}} \leftarrow \theta, \quad k \leftarrow 0$ ▷ copy: $O(p)$
    else
      $k \leftarrow k + 1$
      if $k = P$ **then break**
      end if
    end if
  end for
  return θ_{best}
end function

Running time — early stopping

Overhead per epoch: one validation pass, $O(n_{\text{valid}}\, p)$, and an $O(p)$ copy when the loss improves. **Memory:** one extra copy of the parameters. It usually *saves* time by ending training early. The validation set used to stop cannot also give an unbiased error estimate; that is the job of the test set.

Gradient clipping. One unusual mini-batch can give a huge gradient and throw the parameters far away. *Clipping* [14] rescales the gradient when its norm exceeds a threshold G: if $\|\mathbf{g}\| > G$, replace $\mathbf{g}$ by $G\,\mathbf{g}/\|\mathbf{g}\|$ (optional line of Algorithm 3). For example, $\mathbf{g} = (6, 8)$ with $G = 5$ becomes $(3, 4)$. The direction is kept, only the length is limited; the cost is $O(p)$.

Modern connection — how a large language model is trained

The recipe for GPT-3, typical of large models, uses many of the ideas above [5]: Adam with $\beta_1 = 0.9$ and $\beta_2 = 0.95$, the gradient norm clipped at 1.0 to tame rare large steps, a linear warm-up over the first 375 million tokens followed by cosine decay of the learning rate to 10% of its peak over 260 billion tokens (Section 3.3), weight decay of 0.1, and mini-batches that grow to about 3.2 million tokens. No second-order method, line search or validation-based early stopping is used at this scale. The objective is the next-token cross-entropy (Note 2, Section 2), and each update is a mini-batch gradient step of Section 5, on 1.75×10^{11} parameters. Stored as 32-bit numbers, the parameters and the two Adam moment vectors alone would occupy 3×700 GB $= 2.1$ TB, so the optimiser state is spread over many accelerators; anything quadratic in p, such as a Hessian, is out of the question.

Costs at a glance. Table 1 collects the running times of this note (prediction cost depends only on the model).

Table 1. Cost of the optimisation methods of this note: work per iteration, memory beyond the data, and typical number of iterations (κ: condition number, τ: target accuracy, M: L-BFGS history length).

method	per iteration	memory	iterations and remarks
batch gradient descent	$O(np)$	$O(p)$	$\sim \frac{\kappa}{2} \ln \frac{1}{\tau}$ on a quadratic (best fixed η)
+ line search	$O(np)$ per trial	$O(p)$	no tuning of η; deterministic only
mini-batch SGD	$O(n'p)$	$O(p)$	n/n' updates per epoch; noise floor if η fixed
momentum	gradient $+O(p)$	$O(p)$	$\sim \frac{\sqrt{\kappa}}{2} \ln \frac{1}{\tau}$ on a quadratic
Newton	$O(np^2 + p^3)$	$O(p^2)$	5–20; quadratic convergence; small p only
conjugate gradient	$O(np)$ (one $A\mathbf{d}$)	$O(p)$	$\leq p$; $\sim \frac{\sqrt{\kappa}}{2} \ln \frac{2}{\tau}$ on a quadratic
L-BFGS	$O(np + Mp)$	$O(Mp)$	medium p, deterministic problems
Adam / AdamW	$O(n'p)$	$O(p)$ ($\approx 4p$)	default for deep networks
finite differences	$O(np^2)$	$O(p)$	checking only
reverse-mode AD	$O(np)$	stored activations	how gradients are computed in practice
projected GD	$O(np)$ + projection	$O(p)$	projection $O(p)$ to $O(p \log p)$ for simple sets

Summary The gradient points uphill and vanishes at minima (stationarity); the Hessian describes curvature and classifies stationary points. By Cauchy–Schwarz, $-\nabla J$ is the locally steepest direction, and gradient descent steps along it with learning rate η. On a quadratic each eigen-direction is multiplied by $1 - \eta\lambda_i$ per step, so stability needs $\eta < 2/\lambda_{\max}$ and the speed is limited by the condition number κ; standardised features help. Convex objectives have a single basin; neural networks do not, but saddles rather than bad minima are the usual obstacle. Mini-batch gradients are unbiased with variance $\propto 1/n'$: many cheap noisy steps, with a noise floor that a decaying learning rate removes. Momentum cuts the dependence on κ to $\sqrt{\kappa}$ at the cost of one vector, and conjugate gradient achieves the same on quadratics without tuning; Newton's method removes it at $O(p^2)$ memory and $O(p^3)$ time; Adam gives every parameter its own step size. Constraints are handled by reparameterisation or projection, and discrete choices by search on validation data. Reverse-mode automatic differentiation supplies exact gradients at the cost of a few function evaluations, and early stopping on validation data decides when to stop.

Questions and exercises

Questions are sorted by kind. Parts A and B are of the kind used in the midterm examination: they test understanding, interpretation and short calculations (at most simple derivatives), not long proofs. Parts C and D go further and are recommended for practice and for the project.

A. Conceptual questions

A1. Using the first-order Taylor expansion, explain why $-\nabla J(\theta)$ is the steepest descent direction. Why is this only a *local* statement?

A2. True or false? Justify each answer in one sentence. (a) If $\nabla J(\theta) = \mathbf{0}$, then θ is a minimum. (b) At a minimum of a differentiable function on $\mathbb{R}^p$ the gradient is zero. (c) The gradient is parallel to the contour lines of J. (d) Every local minimum of a convex function is global.

A3. What do the Hessian eigenvalues at a stationary point tell you? How does $\det H$ reveal a saddle when $p = 2$?

A4. For batch gradient descent on a quadratic bowl, sketch (i) the path of θ across the valley and (ii) the training loss against the iteration number when the learning rate is (a) much too small, (b) slightly below $2/\lambda_{\max}$, (c) above $2/\lambda_{\max}$. Why does the loss still fall steadily in (b), even though θ jumps across the valley at every step?

A5. Why does the condition number $\kappa = \lambda_{\max}/\lambda_{\min}$, and not the size of the eigenvalues, determine how many steps gradient descent needs? How can standardising the features change κ?

A6. For $J = \frac{1}{2}\theta^{\mathsf{T}} A\theta$ with $A = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$ (eigenvalue 9 along $(1,1)$, eigenvalue 1 along $(1,-1)$), gradient descent with $\eta = 0.2$ gives a larger J than with $\eta = 0.1$ after three steps but a much smaller one after forty. Explain why, using the per-direction factors $1 - \eta\lambda_i$.

A7. What does a learning-rate schedule do that a fixed learning rate cannot? Why do large models start with a warm-up, and why does the learning rate decrease towards the end of training?

A8. What does the Armijo condition of backtracking line search check, and why is line search common with full gradients but rare with mini-batch gradients?

A9. Give one convex and one non-convex objective mentioned in this note (the landscape of Figure 1 counts). Why is convexity useful, and why are saddle points, rather than poor local minima, the usual obstacle in high-dimensional non-convex problems?

A10. Compare batch, mini-batch and stochastic gradient descent: cost and noise of one update, updates per epoch, use of parallel hardware.

A11. The mini-batch gradient is unbiased. Why, then, does SGD with a constant learning rate not converge to the minimiser? What is the role of each Robbins–Monro condition?

A12. Doubling the batch size doubles the cost of a step. By what factor does the standard deviation of the gradient estimate fall (batches drawn with replacement)? Why is the data set shuffled at the start of every epoch?

A13. Explain with a heavy ball in a narrow valley why momentum helps. What happens for $\beta = 0$, and for β very close to 1?

A14. Newton's method minimises any quadratic in one step. Give three reasons why it is not used to train a network with 10^8 parameters, and explain why it can converge to a saddle point.

A15. Why does the first step of Adam move every parameter by about η, whatever the size of its gradient? Give one advantage and one drawback of this behaviour.

A16. What does AdamW change about weight decay compared with adding $\frac{\lambda}{2}\|\theta\|^2$ to the loss in Adam, and why does the difference matter only for adaptive methods?

A17. Why are gradients of large models computed by automatic differentiation and not by finite differences? What are finite differences still used for?

A18. Why can stopping before the training loss reaches its minimum improve the test error? Which data set decides when to stop, and why not the test set? What does the patience P trade off?

A19. What does gradient clipping change in an update, and what does it keep? Why is it useful with mini-batch gradients?

A20. Why can gradient descent not choose the number of neighbours k of k-nearest neighbours, or minimise the 0–1 error of a classifier directly? What is done instead in each case?

A21. Conjugate gradient and tuned momentum both need a number of steps proportional to $\sqrt{\kappa}$ on a quadratic. What does conjugate gradient not need to know in advance, and why is it rarely used with mini-batch gradients?

A22. In projected gradient descent, why does the gradient not have to vanish at the constrained minimum? Give one constraint whose projection is cheap and one way to avoid the constraint by reparameterisation.

B. Short calculations

B1. Let $J(\theta_1, \theta_2) = \theta_1^2 + \theta_1\theta_2 + \theta_2^2 - 3\theta_1$. (a) Compute ∇J and the Hessian. (b) Find the stationary point and classify it. (c) Find the eigenvalues of H, the condition number and the largest stable learning rate. (d) Perform one gradient-descent step from $(0, 0)$ with $\eta = 0.25$.

B2. For $J(\theta) = 2(\theta-1)^2$, write the gradient-descent update in the form $\theta_{t+1} - 1 = a\,(\theta_t - 1)$. Starting from $\theta_0 = 0$, compute three steps for $\eta = 0.1$, 0.4 and 0.6 and describe each behaviour. Which η reaches the minimum in one step, and what is the stability limit?

B3. Find the stationary points of $J(\theta_1, \theta_2) = \theta_1^2 - \theta_2^2 + \frac{1}{2}\theta_2^4$ and classify them with the Hessian. Starting at $(1, 0)$, where does gradient descent with a small step go? And starting at $(1, 10^{-6})$? Explain the difference.

B4. For $J = \frac{1}{2}(\theta_1^2 + 25\,\theta_2^2)$ (Figure 3), substitute $\theta_2 = u/5$. Give the condition numbers and largest stable learning rates before and after, and relate this to standardising features.

B5. Fit $f(x) = wx$ (no intercept) with the loss $L_i = \frac{1}{2}(wx_i - y_i)^2$ to the four examples $(x, y) = (1, 2), (2, 3), (-1, -1), (3, 4)$. At $w = 0$ compute the four per-example gradients, the full gradient, and the mini-batch gradients for $B = \{1, 2\}$ and $B = \{3, 4\}$. Check that the average of these two mini-batch gradients equals the full gradient (the two batches split the data into equal halves, so this is the unbiasedness of Section 5 in miniature), and perform one SGD step with $B = \{1, 2\}$ and $\eta = 0.1$.

B6. On $J(\theta) = (\theta - 3)^2$ from $\theta_0 = 0$, perform two steps of momentum with $\eta = 0.1$ and $\beta = 0.5$ ($v_0 = 0$). Compare θ_2 with two steps of plain gradient descent with the same η.

B7. For $J(\theta) = e^\theta - 2\theta$, compute J' and J'' and the minimiser. Perform two Newton steps from $\theta_0 = 0$ and compare with two gradient-descent steps with $\eta = 0.5$ from the same start. Explain why gradient descent does so well here (hint: compare η with $1/J''(\theta^*)$). You may use $e^{0.5} = 1.649$.

B8. (a) Compute the first Adam step ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\eta = 0.01$, ϵ negligible) for the gradient $\mathbf{g}_1 = (0.01, -40)$ and explain the result. (b) A model has $p = 10^8$ parameters stored as 32-bit floats. How much memory does its Hessian need? Compare with the memory for the parameters and the two moment vectors of Adam (not counting the gradient).

B9. (a) With $\eta_{\max} = 0.2$, compute the $1/t$ decay $\eta_t = \eta_{\max} t_0/(t_0 + t)$ with $t_0 = 50$ at $t = 50$ and $t = 150$, and the warm-up-plus-cosine schedule $\eta_t = \eta_{\max}(t + 1)/W$ for $t < W$, $\eta_t =$

$\tfrac{1}{2}\eta_{\max}\big(1+\cos(\pi(t-W)/(T-W))\big)$ for $W \le t \le T$, with $W = 10$ and $T = 110$, at $t = 0$, 4, 60 and 110. (b) Early stopping with patience P stops once the validation loss has not improved on its best value for P consecutive epochs, and returns the best parameters. The validation losses after epochs $1, \dots, 8$ are 0.80, 0.55, 0.47, 0.49, 0.45, 0.46, 0.48, 0.44. With $P = 2$, after which epoch does training stop, and which epoch's parameters are returned? Would $P = 3$ have changed the answer?

C. Derivations

C1. Let $J(\theta) = \tfrac{1}{2}\theta^{\mathsf{T}} A\theta - \mathbf{b}^{\mathsf{T}}\theta$ with A symmetric positive definite. Derive ∇J, H and θ^{*}. Show that the error of gradient descent obeys $\boldsymbol{\Delta}_{t+1} = (I - \eta A)\boldsymbol{\Delta}_t$, that $\eta = 2/(\lambda_{\max} + \lambda_{\min})$ reduces the error norm by at least $(\kappa - 1)/(\kappa + 1)$ per step, and that Newton's method converges in one step from any start.

C2. Show that the mini-batch gradient is unbiased and that its covariance is $\tfrac{1}{n'}$ times that of a single-example gradient when the batch is drawn with replacement. Without replacement, show that the variance is multiplied by $(n - n')/(n - 1)$ instead (hint: covariance of two different draws).

C3. Apply momentum to $J(\theta) = \tfrac{1}{2}a\theta^2$ with curvature $a > 0$. Show that (θ_{t+1}, θ_t) is obtained from (θ_t, θ_{t-1}) by a 2×2 matrix, and that for $\beta = 0$ the convergence factor is $|1 - \eta a|$. Show that a constant gradient g leads to the limiting velocity $-\eta g/(1 - \beta)$.

C4. Starting from the chord definition of convexity, derive the tangent-plane inequality $J(\mathbf{b}) \ge J(\mathbf{a}) + \nabla J(\mathbf{a})^{\mathsf{T}}(\mathbf{b} - \mathbf{a})$ and conclude that every stationary point of a convex function is a global minimum. Show that the mean of convex losses plus $\lambda \lVert\theta\rVert^2$ ($\lambda \ge 0$) is convex.

C5. For a one-dimensional J with $J'(\theta^{*}) = 0$ and $J''(\theta^{*}) > 0$, expand J' and J'' around θ^{*} to show that the Newton error $\Delta_t = \theta_t - \theta^{*}$ satisfies $\Delta_{t+1} = \dfrac{J'''(\theta^{*})}{2J''(\theta^{*})}\,\Delta_t^2 + O(\Delta_t^3)$. Check the prediction numerically on $J(\theta) = e^{\theta} - 2\theta$.

D. Programming

D1. Implement batch, mini-batch and stochastic gradient descent for logistic regression and reproduce Figure 5. Then give SGD the schedule $\eta_t = \eta_0/(1 + t/t_0)$ and report the excess loss after 20 epochs.

D2. Implement gradient descent ($\eta = 0.1$), momentum ($\eta = 0.1$, $\beta = 0.5$), Newton's method (undamped) and Adam ($\eta = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) for $J = \tfrac{1}{4}\theta_1^4 - \theta_1\theta_2 + \tfrac{1}{2}\theta_2^2$ (Figure 1). Count the iterations each needs to reach $\lVert\nabla J\rVert < 10^{-8}$ from $(2, 1)$. Map the starting points in $[-2, 2]^2$ from which undamped Newton converges to the saddle, and check that Algorithm 5 avoids it.

D3. Write a gradient checker (Algorithm 8) for the logistic-regression loss. Plot the error of forward and central differences against $h \in [10^{-12}, 10^{-1}]$ on log–log axes and explain the curves.

Further reading. Goodfellow, Bengio and Courville [7], chapters 4 and 8; Bottou, Curtis and Nocedal [3]; Nocedal and Wright [13]; Boyd and Vandenberghe [4].

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